

Dashboard Storyboard

EduVision_DV — Global Higher Education Performance Analytics

Project

Infosys Springboard Virtual Internship

Author

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Platform

Tableau Desktop 2026.2 · fixed canvas 1600 × 950 px

Dataset

university_final_dataset.csv — 1,503 institutions × 52 columns

Issued

29 September 2026

Scope of this document. This storyboard is the approved design specification for the four EduVision_DV dashboards: canvas dimensions, colour system, KPI card definitions, chart types and encodings, filter inventory, navigation, and the cross-filter dashboard action. It is the planning artifact that Milestone 3 builds against.

KPI naming. The metric the official specification calls *Research Productivity Index* is implemented here as **Research Productivity (Proxy)**, derived from the THE Research Environment score. It is not Times Higher Education's official Research Productivity metric, and the “(Proxy)” label is carried through every dashboard, legend and tooltip.

1. Executive Summary & Design System

This guide provides the complete blueprint for constructing the 4 interactive dashboards in **Tableau Desktop** using the finalized dataset (`university_final_dataset.csv` or `university_final_dataset.xlsx`).

1.1 Color Palette & Theme Specifications

The visual theme follows a modern, high-contrast dark aesthetic optimized for executive reporting:

Element	Hex Code	Purpose / Visual Role
Canvas Background	#12161A	Deep charcoal dashboard background
Card / Container Fill	#1B2026	Elevated tile background for KPI blocks and charts
Primary Typography	#FFFFFF	Headers, KPI numerals, primary titles
Secondary Typography	#CFD6DF	Subtitles, axis labels, field captions, footers
Gridlines & Borders	#28323D	Subtle divider lines with low visual noise
Primary Accent	#00D2C4	Teal / Cyan (Dashboard 1 & 4 - Overview & Benchmarking)
Research Accent	#FF9F1C	Amber / Orange (Dashboard 2 - Research Analytics)
Student Accent	#2EC4B6	Mint / Emerald (Dashboard 3 - Student Analytics)
Highlight / Alert	#E71D36	Coral Red (Reference lines, maximum highlights)

2. Calculated Fields & Data Prep in Tableau

Before constructing worksheets, create the following calculated fields in Tableau Desktop:

2.1 Display Tooltip Fields

- `Display_Students_per_Staff` : `sql IF ISNULL([KPI_Faculty_to_Student_Ratio]) THEN "Sourced from THE (Data not available)" ELSE STR(ROUND([KPI_Faculty_to_Student_Ratio], 1)) + " students/staff" END`
- `Display_Int_Student_Pct` : `sql IF ISNULL([KPI_International_Student_Pct]) THEN "Sourced from THE (Data not available)" ELSE STR(ROUND([KPI_International_Student_Pct], 1)) + "%" END`
- `Display_Research_Productivity` : `sql IF ISNULL([KPI_Research_Productivity_Proxy]) THEN "THE Environment Proxy (Data not available)" ELSE STR(ROUND([KPI_Research_Productivity_Proxy], 1)) + " pts" END`

2.2 Aggregation & KPI Helper Fields

- `Total_Universities_Count` : `sql COUNTD([Institution_Name])`
- `Avg_Global_Score` : `sql AVG([KPI_Global_Ranking_Score])`
- `Avg_Academic_Reputation` : `sql AVG([KPI_Academic_Reputation_Score])`
- `Avg_Research_Impact` : `sql AVG([KPI_Research_Impact_Score])`
- `Avg_Students_per_Staff` : `sql AVG([KPI_Faculty_to_Student_Ratio])`
- `Avg_Intl_Student_Pct` : `sql AVG([KPI_International_Student_Pct])`
- `Avg_Research_Productivity` : `sql AVG([KPI_Research_Productivity_Proxy])`
- `Total_FTE_Enrollment` : `sql SUM([THE_No_of_FTE_Students])`

3. Dashboard 1: University Overview

Purpose:

Provides macro-level institutional rankings, geographical spread, and structural distributions across 1,503 institutions.

Layout Dimensions:

- Fixed Size: `1600 x 950` pixels (or Automatic responsive).
- Canvas Background: `#12161A` .

Top KPI Summary Cards (Container: 3 Cards across top):

- Top Ranked University:**
- Main Value:** MIT (or Massachusetts Institute of Technology (MIT))

3. **Subtitle:** #1 in World Rankings 2025
4. **Accent:** #00D2C4
5. **Total Universities:**
6. **Calculation:** COUNTD([Institution_Name])
7. **Main Value:** 1,503
8. **Subtitle:** 106 Countries Covered
9. **Average Global Score:**
10. **Calculation:** AVG([KPI_Global_Ranking_Score])
11. **Main Value:** 41.84
12. **Subtitle:** Avg. of Top 600 Scored Institutions (Do NOT hard-code 41.84; use aggregation).

Visualizations:

1. **Chart 1: Top 10 Universities by Global Score**
2. **Type:** Horizontal Bar Chart (descending).
3. **Rows:** Institution_Name (Filtered to Top 10 by AVG(KPI_Global_Ranking_Score)).
4. **Columns:** AVG(KPI_Global_Ranking_Score) (Axis range: 85 to 100).
5. **Color:** Single Accent #00D2C4 .
6. **Labels:** Value on end of bar (ROUND([KPI_Global_Ranking_Score], 1)).
7. **Chart 2: Universities by Region**
8. **Type:** Donut Chart (Dual Axis Pie Chart).
9. **Color / Slice:** Region .
10. **Angle / Size:** COUNTD(Institution_Name) .
11. **Inner Hole:** Shaded #1B2026 with center label displaying Total 1,503 .
12. **Chart 3: Geographic Distribution**
13. **Type:** Filled World Map / Symbol Density Map.
14. **Location:** Geographic role Location (Country).
15. **Size / Color:** COUNTD(Institution_Name) .
16. **Map Theme:** Dark map layer (#12161A).

Interactive Filters:

- Region

(Multiple values dropdown / buttons)
- Location

(Dropdown, set to "Only Relevant Values")
- SIZE

(Category filter: S, M, L, XL)
- FOCUS

(Category filter: FC, FO, CO, SP)

4. Dashboard 2: Research Analytics

Purpose:

Analyzes research productivity, citations per faculty, and institutional research intensity.

Accent Color:

#FF9F1C (Amber / Orange)

Top KPI Cards:

1. **Highest Research Impact:**
2. Institution with highest Citations_per_Faculty_Score (Score: 100.0).
3. **Average Citation Score:**
4. **Calculation:** AVG([KPI_Research_Impact_Score])
5. **Value:** 23.5
6. **Subtitle:** Citations per Faculty (All 1,503)
7. **Average Research Productivity (Proxy):**
8. **Calculation:** AVG([KPI_Research_Productivity_Proxy])
9. **Value:** 61.2
10. **Subtitle:** THE Environment Metric (195 Matched)

Visualizations:

1. **Chart 1: Research Impact vs Research Productivity (Proxy)**
2. **Type:** Scatter Plot.

3. **Columns (X-axis):** KPI_Research_Productivity_Proxy .
4. **Rows (Y-axis):** KPI_Research_Impact_Score .
5. **Color:** Region .
6. **Detail / Marks:** Institution_Name .
7. **Trend Line:** Linear trendline showing correlation between environment and citation output.
8. **Chart 2: Research Intensity Comparison**
9. **Type:** Box Plot.
10. **Columns:** RES. (Ordered: VH , HI , MD , LO).
11. **Rows:** KPI_Research_Impact_Score .
12. **Visual Rationale:** Demonstrates how Citation Scores distribute systematically higher in Very High (VH) vs Medium (MD) institutions.
13. **Chart 3: Top 10 Research Institutions**
14. **Type:** Grouped Bar Chart / Side-by-Side Bars.
15. **Filter:** Top 10 filtered strictly by KPI_Research_Impact_Score .
16. **Measures:** KPI_Research_Impact_Score and KPI_Research_Productivity_Proxy .
17. **Colors:** Impact (#FF9F1C), Productivity (#48CAE4).

Filters & Mandatory Notice:

- Filters: Region , Location (Only Relevant Values).
- **Mandatory Dashboard Footer / Annotation:**

"Research Productivity (Proxy) uses THE Research Environment and is available only for matched QS-THE institutions."

5. Dashboard 3: Student Analytics

Purpose:

Explores learning environment indicators including student-to-faculty ratios, internationalization, and total enrollment volume.

Accent Color:

#2EC4B6 (Teal / Mint)

Top KPI Cards:

1. **Average Students per Staff:**
2. **Calculation:** AVG([KPI_Faculty_to_Student_Ratio])
3. **Value:** 17.4
4. **Label:** Average Students per Staff (Do **NOT** format as 1:17.4).
5. **Average International Student %:**
6. **Calculation:** AVG([KPI_International_Student_Pct])
7. **Value:** 25.5%
8. **Subtitle:** Global Student Mobility
9. **Total FTE Student Headcount:**
10. **Calculation:** SUM([THE_No_of_FTE_Students])
11. **Value:** Displayed as formatted M (e.g. 5.46M across all 195 matched universities).

Visualizations:

1. **Chart 1: Average Students per Staff by Region**
2. **Type:** Horizontal Bar Chart.
3. **Rows:** Region .
4. **Columns:** AVG([KPI_Faculty_to_Student_Ratio]) .
5. **Label:** STR(ROUND(AVG([KPI_Faculty_to_Student_Ratio]), 1)) + " students/staff" .
6. **Chart 2: Internationalization Scatter**
7. **Type:** Scatter Plot.
8. **Columns (X-axis):** KPI_Faculty_to_Student_Ratio (Students per Staff).
9. **Rows (Y-axis):** KPI_International_Student_Pct .
10. **Color:** Region .
11. **Detail:** Institution_Name .
12. **Tooltip:** Uses Display_Students_per_Staff and Display_Int_Student_Pct .
13. **Chart 3: Top 10 Universities by Enrollment**
14. **Type:** Horizontal Bar Chart.

15. **Filter:** Top 10 by THE_No_of_FTE_Students .
16. **Rows:** Institution_Name .
17. **Columns:** THE_No_of_FTE_Students .
18. **Strict Data Handling:** Unmatched institutions have Null enrollment and must remain omitted (never coerced to 0).

Filters:

- Region , Location (Only Relevant Values), SIZE , FOCUS .

6. Dashboard 4: Country Comparison & Benchmarking

Purpose:

Facilitates national-level performance comparisons with dynamic visual cross-filtering.

Accent Color:

#00D2C4

Top KPI Cards:

1. **Top Country by Average QS Global Score:**
2. Country with highest mean score among institutions with valid KPI_Global_Ranking_Score .
3. **Total Ranked Countries:**
4. **Calculation:** COUNTD([Location])
5. **Value:** 106
6. **Most Represented Country:**
7. **Value:** United States
8. **Subtitle:** 197 Ranked Universities

Visualizations:

1. **Chart 1: Country Benchmarking Heatmap / Summary Table**
2. **Rows:** Location (Country).
3. **Columns (Measures):**

1. AVG([KPI_Global_Ranking_Score])

2. AVG([KPI_Academic_Reputation_Score])

3. AVG([KPI_Research_Impact_Score])

4. COUNTD([Institution_Name]) (University Count)
4. **Formatting:** Color highlight table / gradient background for quick visual outlier detection.
5. **Chart 2: Geographic Country Map**
6. **Type:** Choropleth World Map.
7. **Dimension:** Location .
8. **Color Measure:** COUNTD([Institution_Name]) or AVG([KPI_Academic_Reputation_Score]) .

Dynamic Dashboard Action (Critical Requirement):

- **Action Name:** Filter_Country_From_Benchmarking
- **Source Sheet:** Country_Benchmarking_Table
- **Target Sheet:** Geographic_Country_Map (and associated KPI cards)
- **Run Action on:** Select
- **Clearing Selection:** Show_all_values
- **Effect:** Clicking any country row in the benchmarking table instantly zooms/highlights the corresponding nation on the world map and filters country performance.

7. Step-by-Step Tableau Desktop Workflow

Follow these exact steps inside **Tableau Desktop**:

1. **Connect to Data:**
2. Launch Tableau Desktop.
3. Under *Connect*, select **Text File** and choose university_final_dataset.csv (or **Microsoft Excel** and choose university_final_dataset.xlsx).
4. Verify that 1,503 rows are loaded.
5. **Verify Data Types:**

6. Ensure Rank_2025_Numeric , KPI_Global_Ranking_Score , KPI_Academic_Reputation_Score , KPI_Faculty_to_Student_Ratio , KPI_International_Student_Pct , KPI_Research_Impact_Score , and KPI_Research_Productivity_Proxy are detected as **Number (decimal)**.
7. Ensure Location has Geographic Role → **Country/Region**.
8. **Create Dashboard Sheets:**
9. Build individual worksheets as detailed in Sections 3 through 6.
10. Apply dark background formatting (Format → Shading → Worksheet: #1B2026 , Font: #FFFFFF).
11. **Assemble Dashboards:**
12. Create 4 separate Dashboard tabs:
 1. University Overview
 2. Research Analytics
 3. Student Analytics
 4. Country Comparison
13. Place horizontal/vertical layout containers.
14. Apply dashboard background color #12161A .
15. **Configure Filters & Interactivity:**
16. Add filters to container headers.
17. Set Location filter to **Only Relevant Values**.
18. Under Dashboard → Actions , add the filter action for Dashboard 4 connecting table to map.
19. **Save & Publish:**
20. Save workbook as EduVision_DV.twbx (Tableau Packaged Workbook).